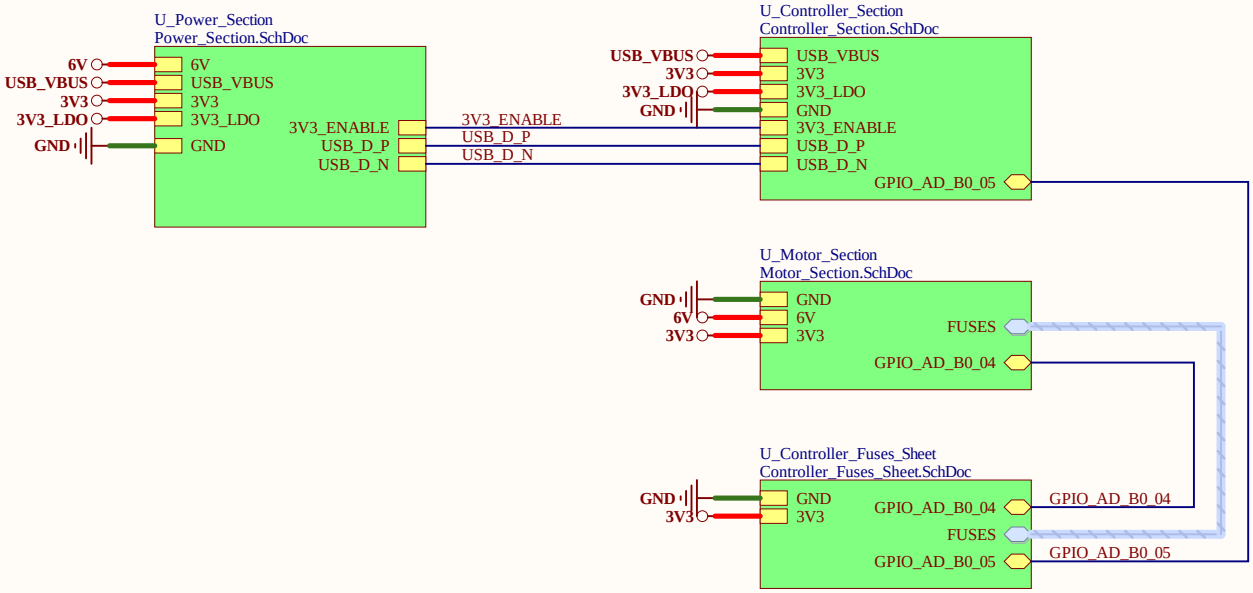
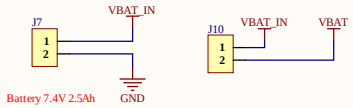


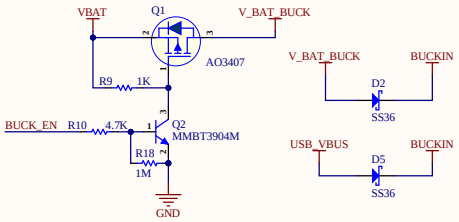
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Title: Top-Sheet.SchDoc	
File: DC-Motor-Driver-v2.0.PrjPcb	
Date: 7/1/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 5

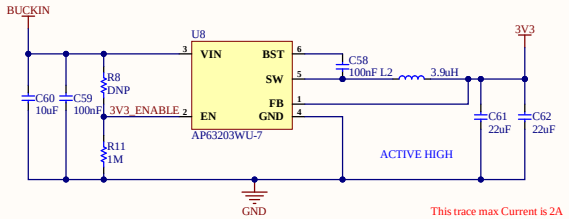
BATTERY



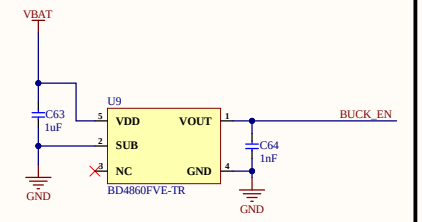
POWER SWITCH FOR BATTERY CUTOFF



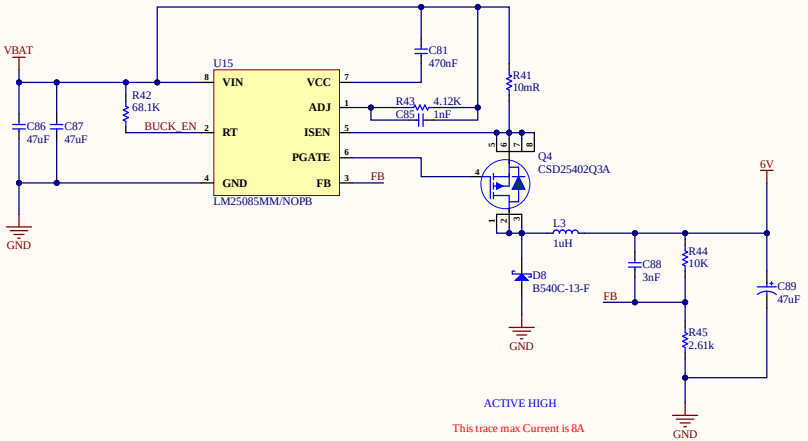
3V3 BUCK REGULATOR



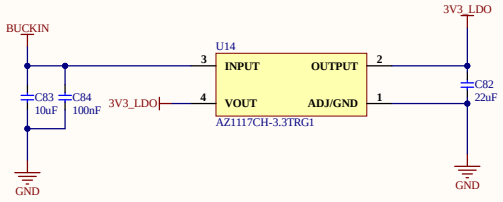
BATTERY SUPERVISORY CIRCUIT



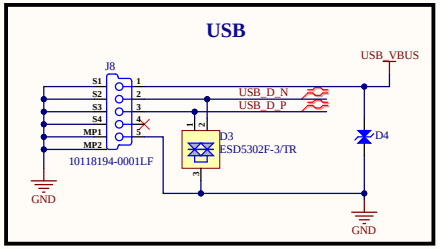
6V BUCK REGULATOR

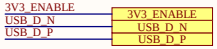


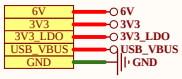
3V3 LDO FOR SNVS



USB

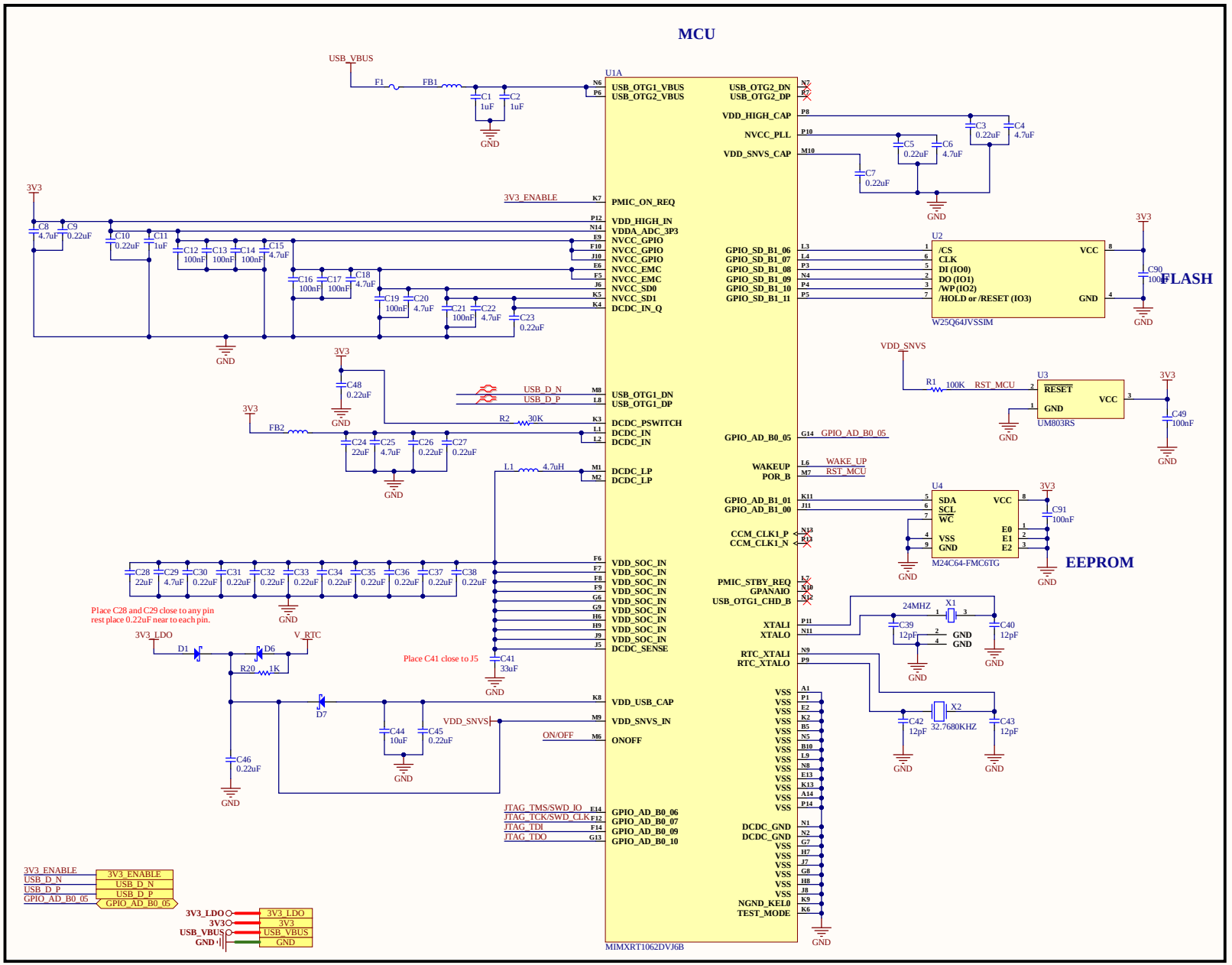




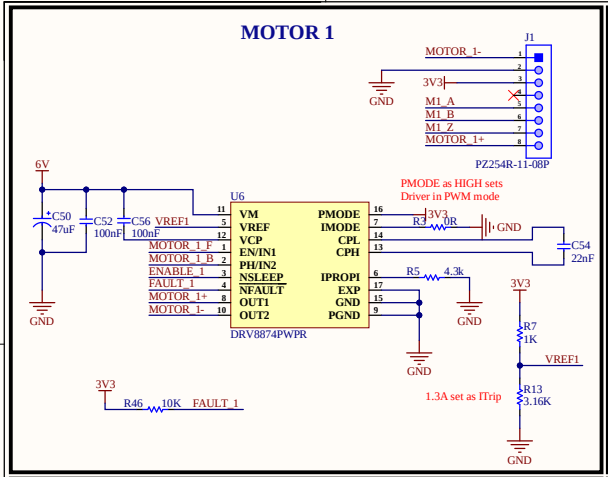


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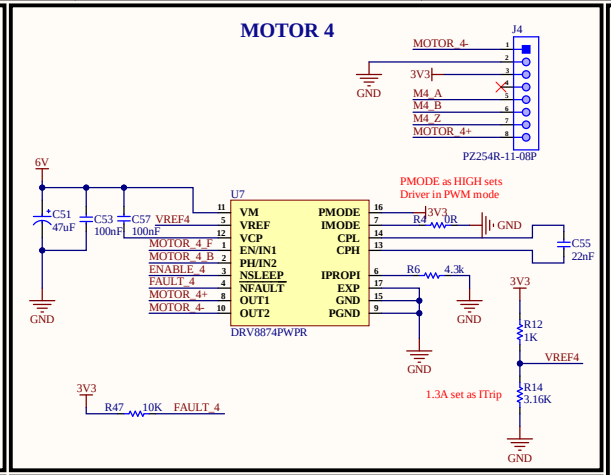
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File: DC-Motor-Driver-v2.0.PrjPcb	
Date: 7/1/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 5


MCU
U1A
MIMXRT1062DVJ6B
USB_VBUS
F1
FB1
C1 1uF
C2 1uF
GND
3V3
3V3_ENABLE
PMIC_ON_REQ
VDD_HIGH_IN
VDDA_ADC_3P3
NVCC_GPIO
NVCC_GPIO
NVCC_EMC
NVCC_EMC
NVCC_SD0
NVCC_SD1
DCDC_IN_Q
C8 4.7uF
C9 0.22uF
C10 0.22uF
C11 1uF
C12 100nF
C13 100nF
C14 100nF
C15 4.7uF
C16 100nF
C17 100nF
C18 4.7uF
C19 100nF
C20 4.7uF
C21 100nF
C22 4.7uF
C23 0.22uF
USB_OTG1_VBUS
USB_OTG2_VBUS
USB_OTG2_DN
USB_OTG2_DP
VDD_HIGH_CAP
NVCC_PLL
VDD_SNVS_CAP
C3 0.22uF
C4 4.7uF
C5 0.22uF
C6 4.7uF
C7 0.22uF
U2
/CS
CLK
DI (IO0)
DO (IO1)
/WP (IO2)
/HOLD or /RESET (IO3)
VCC
GND
W25Q64JVSSIM
FLASH
C90 100nF
GPIO_SD_B1_06
GPIO_SD_B1_07
GPIO_SD_B1_08
GPIO_SD_B1_09
GPIO_SD_B1_10
GPIO_SD_B1_11
USB_D_N
USB_D_P
USB_OTG1_DN
USB_OTG1_DP
C48 0.22uF
FB2
R2 30K
DCDC_PSWITCH
DCDC_IN
DCDC_IN
C24 22uF
C25 4.7uF
C26 0.22uF
C27 0.22uF
L1 4.7uH
DCDC_LP
DCDC_LP
VDD_SNVS
R1 100K
RST_MCU
U3
RESET
GND
VCC
UM803RS
C49 100nF
GPIO_AD_B0_05
GPIO_AD_B0_05
WAKEUP
POR_B
WAKE_UP
RST_MCU
U4
SDA
SCL
WC
VSS
GND
VCC
E0
E1
E2
M24C64-FMC6TG
EEPROM
C91 100nF
GPIO_AD_B1_01
GPIO_AD_B1_00
CCM_CLK1_P
CCM_CLK1_N
C28 22uF
C29 4.7uF
C30 0.22uF
C31 0.22uF
C32 0.22uF
C33 0.22uF
C34 0.22uF
C35 0.22uF
C36 0.22uF
C37 0.22uF
C38 0.22uF
Place C28 and C29 close to any pin
rest place 0.22uF near to each pin.
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
VDD_SOC_IN
DCDC_SENSE
PMIC_STBY_REQ
GPANAIO
USB_OTG1_CHD_B
3V3_LDO
D1
D6
V_RTC
R20 1K
Place C41 close to J5
C41 33uF
24MHZ
X1
C39 12pF
C40 12pF
XTALI
XTALO
RTC_XTALI
RTC_XTALO
X2
C42 12pF
32.7680KHZ
C43 12pF
VDD_USB_CAP
D7
VDD_SNVS
VDD_SNVS_IN
ON/OFF
ONOFF
C44 10uF
C45 0.22uF
C46 0.22uF
JTAG_TMS/SWD_IO
JTAG_TCK/SWD_CLK
JTAG_TDI
JTAG_TDO
GPIO_AD_B0_06
GPIO_AD_B0_07
GPIO_AD_B0_09
GPIO_AD_B0_10
VSS
DCDC_GND
DCDC_GND
NGND_KEL0
TEST_MODE
3V3_ENABLE
USB_D_N
USB_D_P
GPIO_AD_B0_05
3V3_LDO
3V3
USB_VBUS
GND
SWITCHES
ON/OFF
WAKE_UP
RST_MCU
SW1
SW2
SW3
DEBUGGING HEADER
J11
JTAG_TMS/SWD_IO
JTAG_TCK/SWD_CLK
JTAG_TDO
JTAG_TDI
RST_MCU
20021121-00010C4LF
C47 100nF
1.SWD debug is enabled by default
2.Board rework and fuse programming
are needed to support JTAG debug
COIN CELL
V_RTC
TP1
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Template\espark-consulting-group-800x300.jpg.
File does not exist.
Title: Controller_Section.SchDoc
File: DC-Motor-Driver-v2.0.PrjPcb
Date: 7/1/2024
Page Num: 3
Drawn By: Huzaifa Ali
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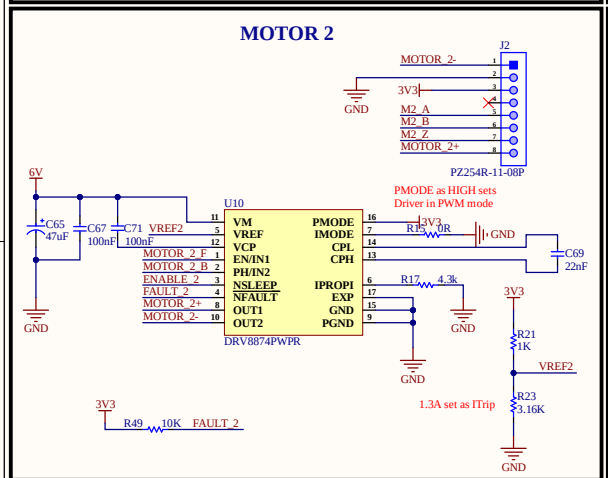
MOTOR 1



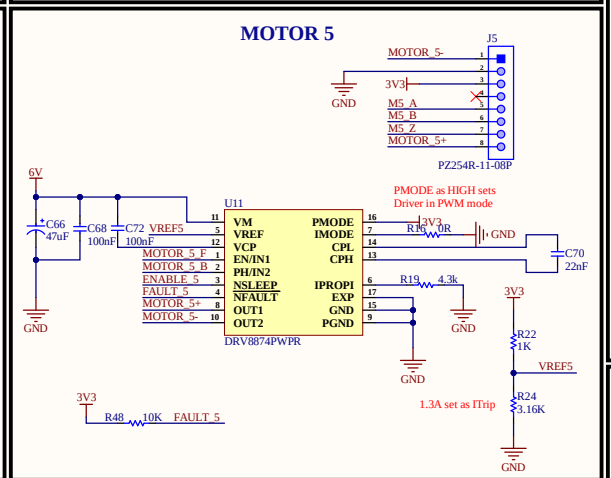
MOTOR 4



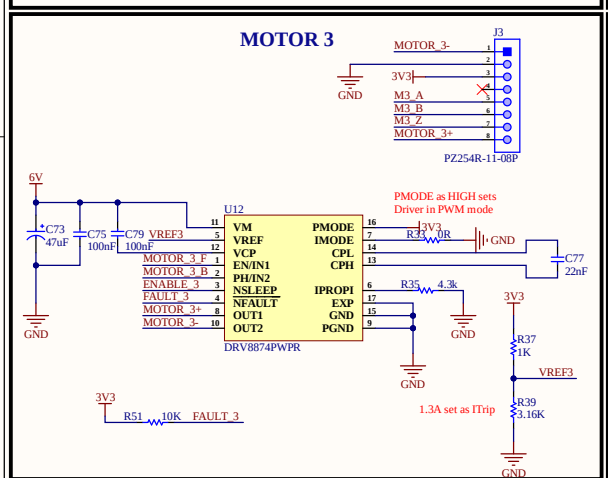
MOTOR 2



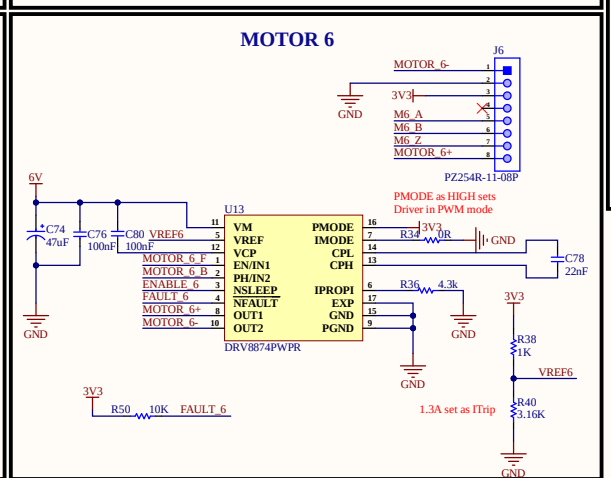
MOTOR 5



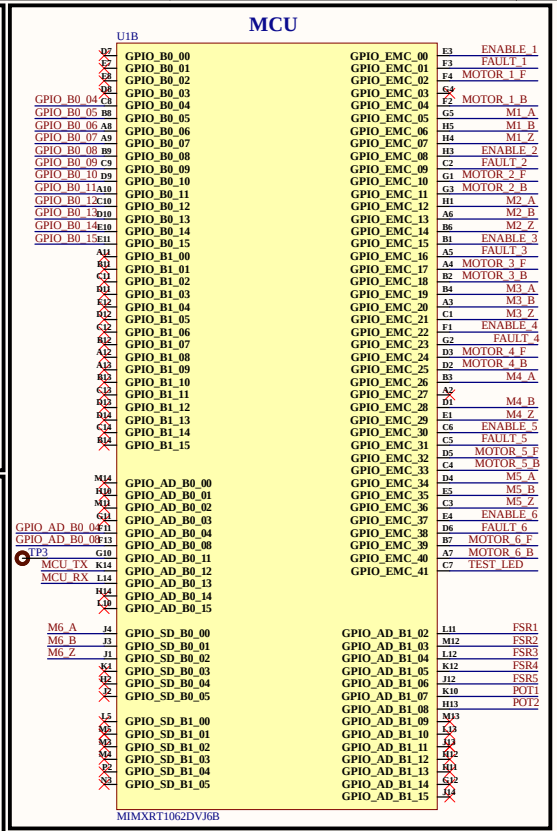
MOTOR 3



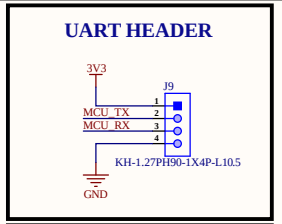
MOTOR 6



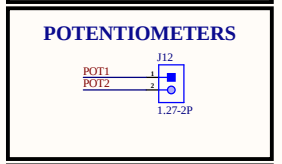
MCU



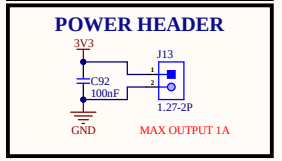
UART HEADER

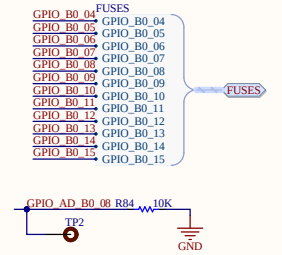


POTENTIOMETERS

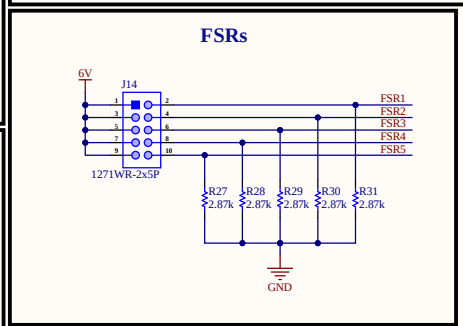


POWER HEADER

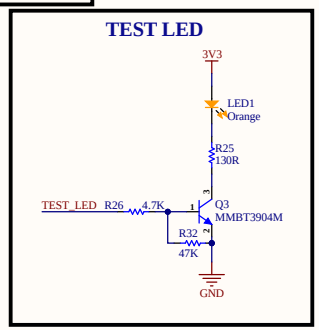


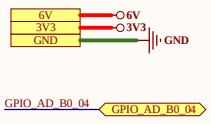


FSRs



TEST LED



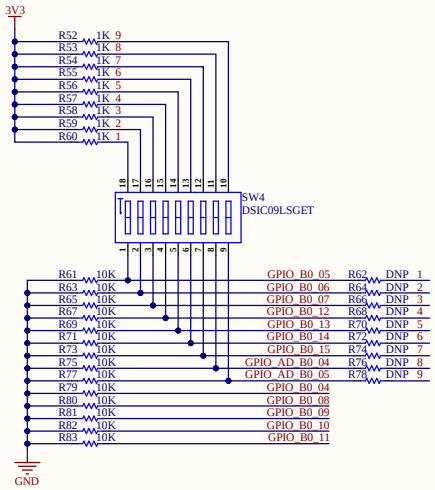


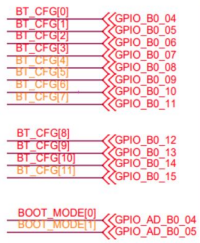
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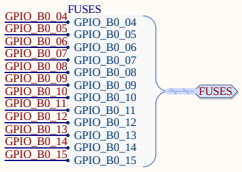
Title: Motor_Section.SchDoc	
File: DC-Motor-Driver-v2.0.PrjPcb	
Date: 7/1/2024	Page Num: 4
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FUSE MAP

TYPE	BOOT_CFG[11]	BOOT_CFG[10]	BOOT_CFG[9]	BOOT_CFG[8]	BOOT_CFG[7]	BOOT_CFG[6]	BOOT_CFG[5]	BOOT_CFG[4]	BOOT_CFG[3]	BOOT_CFG[2]	BOOT_CFG[1]	BOOT_CFG[0]
	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
FlexSPI1 - Serial NOR	Infinit-Loop: (Debug USE only) 0 - Disable 1- Enable	FLASH_TYPE 000-Device supports 3B read by default 001-Device supports 4B read by default 010-HyperFlash 1V8 011-HyperFlash 3V3 100-MXIC Octal DDR			0	0	0	0	HOLD TIME: 00 - 500us 01 - 1ms 10 - 3ms 11 - 10ms		EncryptedXIP 0 - Disabled 1- Enabled	Reserved
SD	Infinit-Loop: (Debug USE only) 0 - Disable 1- Enable	Reserved	Bus Width: 0 - 1-bit 1 - 4-bit	SD1 VOLTAGE SELECTION: 0 - 3.3V 1 - 1.8V	0	1	SD/SDXC Speed: 00 - Normal/SDR12 01 - High/SDR25 10 - SDR50 11 - SDR104		SD Power Cycle Enable: '0' - No power cycle '1' - Enabled via USDHC_RST pad	SD Loopback Clock Source Sel: (for SDR50 and SDR104 only) '0' - through SD '1' - direct	Port Select: 0 - eSDHC1 1 - eSDHC2	Fast Boot: 0 - Regular 1 - Fast Boot











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Title: Controller_Fuses_Sheet.SchDoc	
File: DC-Motor-Driver-v2.0.PrjPcb	
Date: 7/1/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5